Problem 5.1

1) $X(n) = a\,X(n-1) + W(n)$

$E[W(n)W(n+k)] = \sigma_W^2(n)\,\delta(k)$

$\sigma_W^2(n) = \sigma_W^2 = 1 - a^2 = 0{,}36$

2) $Z(n) = X(n) + V(n)$

$E[V(n)V(n+k)] = \sigma_V^2(n)\,\delta(k)$

$\sigma_V^2(n) = \sigma_V^2 = 4$

3) Predict $\hat{X}^-(n) = a\,\hat{X}(n-1)$

$\hat{Z}^-(n) = \hat{X}^-(n)$

$P^-(n) = a^2 P(n) + \sigma_W^2$

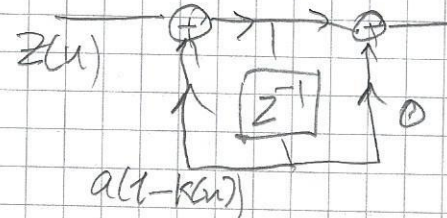
Update $\hat{X}(n) = \hat{X}^-(n) + K(n)(Z(n) - \hat{Z}^-(n))$

$K(n) = \dfrac{P^-(n)}{P^-(n) + \sigma_V^2}$

$P(n) = (1 - K(n))\,P^-(n)$

4) $\hat{X}(n) = K(n)\,Z(n) + a(1-K(n))\,\hat{X}(n-1)$

5)



6) $n=0$ $\hat{X}(0) = E[X(0)] = 0$

$P(0) = V[X(0)] = 1$

$n=1$ $\hat{X}^-(1) = a\,\hat{X}(0) = 0$

$P^-(1) = a^2 P(0) + \sigma_W^2 = 1$

$K(1) = 1/5$

$\hat{X}(1) = 0{,}2\,Z(1)$

$P(1) = 0{,}8$

$n=2$ $\hat{X}^-(2) = 0{,}16\,Z(1)$

$P^-(2) = 0{,}8720$

$K(2) = 0{,}1790$

$\hat{X}(2) = 0{,}1314\,Z(1) + 0{,}1790\,Z(2)$

$P(2) = 0{,}7159$

7) see m-file

8) $$P_\infty = \frac{\sigma_v^2 (a^2 P_\infty + \sigma_w^2)}{a^2 P_\infty + \sigma_w^2 + \sigma_v^2}$$

$$\Downarrow$$

$$0{,}64 P_\infty^2 + 1{,}8 P_\infty - 1{,}44 = 0$$

$$\Downarrow$$

$$P_\infty = 0{,}6498$$

9) $\bar{P_\infty} = 0{,}7759$

$K_\infty = 0{,}1625$

Note $\bar{P_\infty} > P_\infty$

10) see m-file

11) $$\hat{x}(n) = K_\infty z(n) + a(1-K_\infty)\hat{x}(n-1)$$

$$H(z) = \frac{K_\infty}{1 - a(1-K_\infty)\bar{z}^1} = \frac{0{,}1625}{1 - 0{,}6700\,\bar{z}^1}$$

12) KF better than non-causal IIR Wiener

KF equal to causal —"—

13)
14)
15) see m-file

Problem 5.2

see m-file